

Mikhail Razakov

Research Software Engineer | C++ · Python · Algorithms

Computer science and algorithmic problem-solving background with hands-on C++ and Python, compiler engineering, and independent research-oriented systems work. Built and tested an LLVM optimization, graph algorithms, and program-analysis tools with an emphasis on correctness, reproducibility, and adversarial cases. Interested in applying these skills to quantitative research and systematic equity strategies.

EXPERIENCE

? MCST ? compiler engineering intern · July 1 ? August 31, 2026 · LLVM · C++23 · Python · 0.25 FTE. Implemented an LLVM loop-invariant code-motion pass in C++; reasoned about loop invariance, side effects, memory access, speculative safety, and loop structure. Built a Python verification harness comparing behavior before and after optimization and checking transformation / no-transformation cases. Implemented and tested RPO with cycle detection, Dijkstra, Dinic max-flow, and Tarjan SCC.

? UniversalToolchain / Wist2 ? creator and primary developer · 2024 ? present. Independently designed a compiler/language system with intermediate representations, SSA/optimizations, and interpreter/CIL backends. Use benchmarking, differential/metamorphic verification, and reproducible checks. Verification as of July 25, 2026: 1,465/1,465 tests with zero failures across nine packages and clean-consumer projects.

? Mama Lama ? Software Developer · 2023 ? 2025. Independently took event applications from requirements and architecture through implementation, testing, and finished products.

PROJECTS

? PS-form Memory Dependence Analyzer ? C17 program analysis. Ranked 1st of 49; only 5.0/5.0 result; 104/104 official tests. Exact oracle on small domains, randomized/metamorphic testing, stress testing, and reproducible WA/TL counterexamples.

? UniversalToolchain / Wist2 ? .NET compiler/language platform. Benchmarking, differential/metamorphic verification, SSA/optimizations, and interpreter/CIL parity.

? x86-64 Codegen & Register Allocation Lab ? Rust, SSA-like IR, liveness, linear scan, and differential validation.

SKILLS & RECOGNITION

Programming: C++23, Python, C17, C#, Rust. Algorithms & CS: data structures, graph algorithms, program analysis, CFG/SSA, compiler optimizations. Research & verification: exact oracles, differential/metamorphic testing, randomized testing, stress testing, performance experiments, counterexample analysis. Tools: LLVM, Linux, Git, CMake, clang-tidy, ASan/UBSan. Recognition: HSE Vysshaya Proba and regional informatics olympiad prize-winner; first-degree MEPhI Junior diplomas in 2025 and 2026; first-degree Baltic Science and Engineering Competition diploma and Excellence as Hope prize.

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